Unit Functional Design Specification – RO Plant 1 & 2

Wilmar Industrial Complex PT MNA Serang, Indonesia

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1 EQUIPMENT MODULE CLASS DESIGN – _EM_WTP_FILT_SYS

1.1 Functional Description

EM_WTP_FILT_SYS is a class-based equipment module that drives three other sequences to complete the filtration system for different products as per the selected jobs. EM_WTP_FILT_SYS is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. All devices used in EM shall be acquired in Service Command and It shall be released in Backwash & Rinsing Command. Hold Request shall be enabled for all acquired devices. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as buttons on graphics wherever possible.

1.2 Failure Monitors

Fail Monitor Name	Condition
HM01	Module Ownership Lost
HM02	Raw Water Tank Level LT1501 is in low level
HM03	Filtered Water Tank Level LT1523 is in high level
HM04	Incoming Raw Water Flowrate FT1501 detect no flow
HM05	PAC/Coagulant Tank Level Switch LSLL1501 low alarm
HM06	NaOCl Tank Level Switch LSLL1502 low alarm
HM07	RO Concentrate Storage Tank LT1542 is in low level
HM08	Backwash Water Flowrate FT1524 detect no flow
HM09	Raw Water Booster Pumps Tripped
HM10	Backwash Pumps Tripped
HM11	Dosing Pumps Tripped
HM12	5 hours of Sequence Timer
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.

Fail Monitor Name	Condition
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Device 15 Failed.
SM16	Device 16 Failed.
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.
SM20	Device 20 Failed.
SM21	Device 21 Failed.
SM22	Device 22 Failed.
SM23	Device 23 Failed.
SM24	Device 24 Failed.
SM25	Device 25 Failed.
SM26	Device 26 Failed.
SM27	Device 27 Failed.
SM28	Device 28 Failed.
SM29	Device 29 Failed.

Table 1: Failure Monitors

1.3 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	_EM_WTP_FILT_SYS	EM_WTP_FILT_A
DEV01	_V12_IP_130_SE	WP1_V1521A
DEV02	_V12_IP_130_SE	WP1_V1522A
DEV03	_V12_IP_130_SE	WP1_V1523A
DEV04	_V12_IP_130_SE	WP1_V1524A
DEV05	_V12_IP_130_SE	WP1_V1525A
DEV06	_V12_IP_130_SE	WP1_V1521B
DEV07	_V12_IP_130_SE	WP1_V1522B
DEV08	_V12_IP_130_SE	WP1_V1523B
DEV09	_V12_IP_130_SE	WP1_V1524B
DEV10	_V12_IP_130_SE	WP1_V1525B
DEV11	_MVS11_IL_130_SO	WP1_PU1501A
DEV12	_MVS11_IL_130_SO	WP1_PU1501C
DEV13	_MRSS11_130_SOFT	WP1_PU1502A
DEV14	_MRSS11_130_SOFT	WP1_PU1502B
DEV15	_MRSS11_130_SE	WP1_PU1503A
DEV16	_MRSS11_130_SE	WP1_PU1503B
DEV17	_MRSS11_130_SE	WP1_PU1504A
DEV18	_MRSS11_130_SE	WP1_PU1504B
DEV19	_AI_130_SE	WP1_PD1521
DEV20	_V12_IP_130_SE	WP1_V1521C
DEV21	_V12_IP_130_SE	WP1_V1522C
DEV22	_V12_IP_130_SE	WP1_V1523C
DEV23	_V12_IP_130_SE	WP1_V1524C
DEV24	_V12_IP_130_SE	WP1_V1525C
DEV25	_V12_IP_130_SE	WP1_V1521D
DEV26	_V12_IP_130_SE	WP1_V1522D
DEV27	_V12_IP_130_SE	WP1_V1523D
DEV28	_V12_IP_130_SE	WP1_V1524D
DEV29	_V12_IP_130_SE	WP1_V1525D

Table 2: Instances for Train A

Device Name	Module Class	Instance Name
EM	_EM_WTP_FILT_SYS	EM_WTP_FILT_B
DEV01	_V12_IP_130_SE	WP1_V1521E
DEV02	_V12_IP_130_SE	WP1_V1522E
DEV03	_V12_IP_130_SE	WP1_V1523E
DEV04	_V12_IP_130_SE	WP1_V1524E
DEV05	_V12_IP_130_SE	WP1_V1525E
DEV06	_V12_IP_130_SE	WP1_V1521F
DEV07	_V12_IP_130_SE	WP1_V1522F
DEV08	_V12_IP_130_SE	WP1_V1523F
DEV09	_V12_IP_130_SE	WP1_V1524F
DEV10	_V12_IP_130_SE	WP1_V1525F
DEV11	_MVS11_IL_130_SO	WP1_PU1501B
DEV12	_MVS11_IL_130_SO	WP1_PU1501C
DEV13	_MRSS11_130_SOFT	WP1_PU1502A
DEV14	_MRSS11_130_SOFT	WP1_PU1502B
DEV15	_MRSS11_130_SE	WP1_PU1503A
DEV16	_MRSS11_130_SE	WP1_PU1503B
DEV17	_MRSS11_130_SE	WP1_PU1504A
DEV18	_MRSS11_130_SE	WP1_PU1504B
DEV19	_AI_130_SE	WP1_PD1522
DEV20	_V12_IP_130_SE	WP1_V1521G
DEV21	_V12_IP_130_SE	WP1_V1522G
DEV22	_V12_IP_130_SE	WP1_V1523G
DEV23	_V12_IP_130_SE	WP1_V1524G
DEV24	_V12_IP_130_SE	WP1_V1525G
DEV25	_V12_IP_130_SE	WP1_V1521H
DEV26	_V12_IP_130_SE	WP1_V1522H
DEV27	_V12_IP_130_SE	WP1_V1523H
DEV28	_V12_IP_130_SE	WP1_V1524H
DEV29	_V12_IP_130_SE	WP1_V1525H

Table 3: Instances for Train B

1.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

1.4.1 Command 00 – Hold

Step / Transition No.	Description	Step / Transition Name	Functional Description
H0010	Common Hold Start	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions: Enable Commands
T_H1000	Actions Complete		Go to Step H1000
H1000	Hold Command 1	A010	Hold all Timers & Stop Pumps during HOLD
		A020	Close Valves during HOLD
		EXIT	Enable Valid Command
T_H9900	Actions Complete		Go to Step H9900
H9900	Common Hold End	A010	Release Shared Devices
		EXIT	Hold Logic Complete Actions
HOLD_END	Common Hold End		Hold Complete

Table 4: Hold Sequence

1.4.2 Command 01 – Service

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0000	Start	A010	Enable/Disable Report Parameters and Operational Parameters
T0010	First Pass Check		First Pass = False then S0010
T0020A	First Pass and Task Pointer Check		First Pass = True and Task Pointer != 99 and then S0020
T9900A	First Pass and Task Pointer Check		First Pass = True and Task Pointer = 99 then S9900
S0010	Initialize	A009	Initialize EM
		A010	Initialize Timers
		A020	Set First Pass Flag as TRUE and Continue Next Task
T0020	Actions Complete		Go to Step S0020
S0020	Acquire Device	A010	Reset Hold Request and Enable Monitors
		A020	Acquired Shared Devices
		A030	Change Mode Devices to AUTO/CAS
T0040	Acquire Complete		Go to Step S0040
T0030	Acquire Failed or Timed Out		Go to Step S0030
T0050	Both PU1501A and PU1501B Fail		Go to Step S0050
S0030	Acquire Failed	A010	Operator Prompt : Acquisition failed for “Device Names”
T0010B	Operator Responded		Go to Step S0010
S0040	Start Service	A010	Start Service
		A020	Open Service Sand & Carbon Valves
		A030	Start Raw Water Booster Pump
		A040	Start Dosing Coagulant Pump and Dosing NaOCl Pump
		A050	Start Service Timer TMR1

Step / Transition No.	Description	Step / Transition Name	Functional Description
		EXIT	Continue Next. Set Task Pointer as 2
		B010	Display Service Remaining Timer TMR1
T9900	DEV19>1 OR TMR1=Complete OR OP002≠0		Go to Step S9900
S0050	Fail	A010	Both PU1501A and PU1501B Failed
FAIL	Command to Start Standby Pump Failed		TRUE
S9900	Service Complete	A010	Stop all Pump
		A020	Close all Valves
		A030	Disable all HM & SM
		A040	Release Shared Device
		EXIT	Completed
END	Command Complete		TRUE

Table 5: Service Sequence

1.4.3 Command 02 – Backwash & Rinsing

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0000	Start	A010	Enable/Disable Report Parameters and Operational Parameters
T0010	First Pass Check		First Pass = False then S0010
T0011	Backwash & Rinsing can only run by ONE Train		Go to Step S0011
T0020A	First Pass and Task Pointer Check		First Pass = True and Task Pointer != 99 then S0020
T9900A	First Pass and Task Pointer Check		First Pass = True and Task Pointer = 99 then S9900
S0010	Initialize	A010	Initialize EM
		A020	Set First Pass Flag as TRUE and Set Task Pointer
T0019	Initialize Complete		Go to Step S0019
S0011	Backwash Fail	A010	Backwash & Rinsing Failed
FAIL	Command to Start Backwash & Rinsing Failed		TRUE
S0019	Pump Check	A010	Stanby Pump Check
T0020	Actions Complete		Go to Step S0020
T0011A	Command FAIL		Go to Step S0011
S0020	Acquire Device	A010	Reset Hold Request and Enable Monitors

Step / Transition No.	Description	Step / Transition Name	Functional Description
		A020	Acquired Shared Devices
		A030	Change Mode Devices to AUTO/CAS
T0030	Acquire Failed or Timed Out		Go to Step S0030
T0040	Acquire Complete AND OP002 check		Go to Step S0040
T0050A	OP002 = 2 OR Task Pointer = 3		Go to Step S0050
T0060A	OP002 = 3 OR Task Pointer = 4		Go to Step S0060
T0070A	OP002 = 4 OR Task Pointer = 5		Go to Step S0070
T0080A	OP002 = 5 OR Task Pointer = 6		Go to Step S0080
T0090A	OP002 = 6 OR Task Pointer = 7		Go to Step S0090
T0100A	OP002 = 7 OR Task Pointer = 8		Go to Step S0100
T0110A	OP002 = 8 OR Task Pointer = 9		Go to Step S0110
S0030	Acquire Failed	A010	Operator Prompt : Acquisition failed for “Device Names”
T0010B	Operator Responded		Go to Step S0010
S0040	Start Sand Backwash SET 1	A010	Start Sand Backwash
		A020	Open Sand Backwash Valves
		A030	Start Filter Backwash Pump
		A031	Start Sand Backwash Timer TMR2
		A040	Stop Filter Backwash Pump
		A050	Close Sand Backwash Valves
		EXIT	Continue Next. Set Task Pointer as 3

Step / Transition No.	Description	Step / Transition Name	Functional Description
		B010	Display Sand Backwash Timer TMR2
T0050	Actions & TMR2 Complete		Go to Step S0050
T9900B	OP002 = 1		Go to Last Step S9900
S0050	Start Sand Rinsing SET 1	A010	Start Sand Rinsing
		A020	Open Sand Rinsing Valves DEV1 and DEV4
		A030	Start Raw Water Booster Pump DEV11 or Standby Pump DEV12
		A031	Start Sand Rinsing Timer TMR3
		A040	Stop Raw Water Booster Pump DEV11 or Standby Pump DEV12
		A050	Close Sand Rinsing Valves DEV1 and DEV4
		EXIT	Continue Next. Set Task Pointer as 4
		B010	Display Sand Rinsing Timer TMR3
T0060	Actions & TMR3 Complete		Go to Step S0060
T9900C	OP002 = 2		Go to Last Step S9900
S0060	Start Carbon Backwash SET 1	A010	Start Carbon Backwash
		A020	Open Carbon Backwash Valves
		A030	Start Filter Backwash Pump
		A031	Start Carbon Backwash Timer TMR4
		A040	Stop Filter Backwash Pump
		A050	Close Carbon Backwash Valves
		EXIT	Continue Next. Set Task Pointer as 5
		B010	Display Service Remaining Timer TMR4
T0070	Actions & TMR4 Complete		Go to Step S0070

Step / Transition No.	Description	Step / Transition Name	Functional Description
T9900D	OP002 = 3		Go to Last Step S9900
S0070	Start Carbon Rinsing SET 1	A010	Start Carbon Rinsing
		A020	Open Carbon Rinsing Valves AND Sand Service Valves
		A030	Start Raw Water Booster Pump
		A031	Start Carbon Rinsing Timer TMR5
		A040	Stop Raw Water Booster Pump
		A050	Close Carbon Rinsing Valves AND Sand Service Valves
		EXIT	Continue Next. Set Task Pointer as 6
		B010	Display Service Remaining Timer TMR5
T0080	Actions & TMR5 Complete		Go to Step S0080
T9900E	OP002 = 4		Go to Last Step S9900
S0080	Start Sand Backwash SET 2	A010	Start Sand Backwash
		A020	Open Sand Backwash Valves
		A030	Start Filter Backwash Pump
		A031	Start Sand Backwash Timer TMR6
		A040	Stop Filter Backwash Pump
		A050	Close Sand Backwash Valves
		EXIT	Continue Next. Set Task Pointer as 7
		B010	Display Service Remaining Timer TMR6
T0090	Actions & TMR6 Complete		Go to Step S0090
T9900F	OP002 = 5		Go to Last Step S9900

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0090	Start Sand Rinsing SET 2	A010	Start Sand Rinsing
		A020	Open Sand Rinsing Valves
		A030	Start Raw Water Booster Pump
		A031	Start Sand Rinsing Timer TMR7
		A040	Stop Raw Water Booster Pump
		A050	Close Sand Rinsing Valves
		EXIT	Continue Next. Set Task Pointer as 8
		B010	Display Service Remaining Timer TMR7
T0100	Actions & TMR7 Complete		Go to Step S0100
T9900G	OP002 = 6		Go to Last Step S9900
S0100	Start Carbon Backwash SET 2	A010	Start Carbon Backwash
		A020	Open Carbon Backwash Valves
		A030	Start Filter Backwash Pump
		A031	Start Carbon Backwash Timer TMR8
		A040	Stop Filter Backwash Pump
		A050	Close Carbon Backwash Valves
		EXIT	Continue Next. Set Task Pointer as 9
		B010	Display Service Remaining Timer TMR8
T0110	Actions & TMR8 Complete		Go to Step S0110
T9900H	OP002 = 7		Go to Last Step S9900
S0110	Start Carbon Rinsing SET 2	A010	Start Carbon Rinsing
		A020	Open Carbon Rinsing Valves AND Sand Service Valve
		A030	Start Raw Water Booster Pump
		A031	Start Carbon Rinsing Timer TMR9
		A040	Stop Raw Water Booster Pump

Step / Transition No.	Description	Step / Transition Name	Functional Description
		A050	Close Carbon Rinsing Valves AND Sand Service Valve
		EXIT	Continue Next. Set Task Pointer as 99
		B010	Display Service Remaining Timer TMR9
T9900	Actions & TMR9 Complete		Go to Step S9900
S9900	COMPLETE	A010	Stop ALL Pump
		A020	Close ALL Valves
		A030	Disable all HM & SM
		A040	Release Shared Device
		A050	Timers Reset
		EXIT	Msg : Backwash & Rinsing Command Complete
END	Command Complete		TRUE

Table 6: Backwash & Rinsing Sequence

1.5 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Reporting Parameters				
RP001	Backwash Queue	Boolean	NA	False
RP002	PU1501C	Boolean	NA	False
RP003	PU1502A	Boolean	NA	False
RP004	PU1502B	Boolean	NA	False
RP005	PU1503A	Boolean	NA	False
RP006	PU1503B	Boolean	NA	False
RP007	PU1504A	Boolean	NA	False
RP008	PU1504B	Boolean	NA	False
RP009	In Service	8-bit Unsigned Integer	NA	0
RP010	Last Sequence	8-bit Unsigned Integer	NA	1
Operating Parameters				
OP001	Start/Stop/Pause	8-bit Unsigned Integer	NA	0
OP002	Individual Backwash	8-bit Unsigned Integer	NA	0
OP003	BW 0-Stop 1-Pause	8-bit Unsigned Integer	NA	2
Tuning Parameters				
N/A	N/A	N/A	N/A	N/A
Timers				
TMR1	Service Timer	Continuous	Hour	24
TMR2	Sand Backwash SET 1	Time-out	Minute	10
TMR3	Sand Rinsing SET 1	Time-out	Minute	3
TMR4	Carbon Backwash SET 1	Time-out	Minute	10
TMR5	Carbon Rinsing SET 1	Time-out	Minute	3
TMR6	Sand Backwash SET 2	Time-out	Minute	10

Parameter	Description	Type	Units	Default Value / Reference
TMR7	Sand Rinsing SET 2	Floating Point	Minute	3
TMR8	Carbon Backwash SET 2	Floating Point	Minute	10
TMR9	Carbon Rinsing SET 2	Floating Point	Minute	3

Table 7: Timers, Reporting, Operating and Tuning Parameters

1.6 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value / Reference
MY_TAG	String	EM_WTP_FILT_A
MY_TAG	String	EM_WTP_FILT_B
OPR_CMD_MASK	32 bit unsigned integer	2047

Table 8: Configurable Parameters

1.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_WTP_FILT	Water Treatment Plant Area: Startup EM

Table 9: Named Set Details

Details of states for NS_WTP_FILT named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Service
02	Backwash and Rinsing

Table 10: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – _EM_WTP_UF_SYS

2.1 Functional Description

EM_WTP_UF_SYS is a class-based equipment module that drives three other sequences to complete the filtration system for different products as per the selected jobs. EM_WTP_UF_SYS is derived from the Command Driven equipment module class _EM_CMD16_130 which is defined in a separate document. All devices used in EM shall be acquired in “Service” Command and It shall be released in “Backwash” Command. Hold Request shall be enabled for all acquired devices. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as buttons on graphics wherever possible.

2.2 Failure Monitors

The following Table contains the failure monitors of the equipment module class.

Fail Monitor Name	Condition
HM01	Module Ownership Lost
HM02	Filtered Water Tank Level LT1523 is in low level
HM03	UF Filtrate Tank Level LT1534 is in high level
HM04	Flow Element (FT1535A/FT1535B) detect no flow
HM05	UF System Pressure Switch PSH1531A/PSH1531B high alarm
HM06	Flow Switch FSL1501 detect no flow
HM07	UF Feed Pumps Tripped
HM08	UF Backwash Pumps Tripped
HM09	UF Filtrate Tank LT1534 is in low level
HM10	Citric Acid Level Switch LSLL1533 low alarm
HM11	NaOCl Level Switch LSLL1534 low alarm
HM12	4 hours of Sequence Timer
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.

Fail Monitor Name	Condition
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.

Table 11: Failure Monitors

2.3 Instances

The following Table contains the module instances of the equipment module class.

Instances for Train A

Device Name	Module Class	Instance Name
EM	EM_WTP_UF_SYS	EM_WTP_UF_SYS_A
DEV01	_V12_IP_130_SE	WP1_V1531A
DEV02	_V12_IP_130_SE	WP1_V1532A
DEV03	_V12_IP_130_SE	WP1_V1533A
DEV04	_V12_IP_130_SE	WP1_V1534A
DEV05	_V12_IP_130_SE	WP1_V1535A
DEV06	_V12_IP_130_SE	WP1_V1536A
DEV07	_V12_IP_130_SE	WP1_V1537A
DEV08	_MVS11_IL_130_SO	WP1_PU1535A
DEV09	_MVS11_IL_130_SO	WP1_PU1535B
DEV10	_MRSS11_130_SE	WP1_PU1536A
DEV11	_MRSS11_130_SE	WP1_PU1536B
DEV12	_MRSS11_130_SE	WP1_PU1537
DEV13	_MRSS11_130_SE	WP1_PU1538

Table 12: Instances for Train A

Instances for Train B

Device Name	Module Class	Instance Name
EM	EM_WTP_UF_SYS	EM_WTP_UF_SYS_B
DEV01	_V12_IP_130_SE	WP1_V1531B
DEV02	_V12_IP_130_SE	WP1_V1532B
DEV03	_V12_IP_130_SE	WP1_V1533B
DEV04	_V12_IP_130_SE	WP1_V1534B
DEV05	_V12_IP_130_SE	WP1_V1535B
DEV06	_V12_IP_130_SE	WP1_V1536B

Device Name	Module Class	Instance Name
DEV07	_V12_IP_130_SE	WP1_V1537B
DEV08	_MVS11_IL_130_SO	WP1_PU1535C
DEV09	_MVS11_IL_130_SO	WP1_PU1535B
DEV10	_MRSS11_130_SE	WP1_PU1536A
DEV11	_MRSS11_130_SE	WP1_PU1536B
DEV12	_MRSS11_130_SE	WP1_PU1537
DEV13	_MRSS11_130_SE	WP1_PU1538

Table 13: Instances for Train B

2.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.4.1 Command 00 – Hold

Step / Transition No.	Description	Step / Transition Name	Functional Description
H0010	Common Hold Start	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions: Enable Commands
T_H1000	Actions Complete		Go to Step H1000
H1000	Hold Command 1	A010	Hold all Timers & Stop Pumps during HOLD
		A020	Close Valves during HOLD
		EXIT	Enable Valid Command
T_H9900	Actions Complete		Go to Step H9900
H9900	Common Hold End	A010	Release Shared Devices
		EXIT	Hold Logic Complete Actions
HOLD_END	Common Hold End		Hold Complete

Table 14: Hold Sequence

2.4.2 Command 00 –Service

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0000	Start	A010	Enable/Disable Report Parameters and Operational Parameters
T0010	First Pass Check		First Pass = False then S0010
T0020A	First Pass and Task Pointer Check		First Pass = True and Task Pointer != 99 and then S0020
T9900A	First Pass and Task Pointer Check		First Pass = True and Task Pointer = 99 then S9900
S0010	Initialize	A010	Initialize EM
		A020	Initialize Timers
		A030	Set First Pass Flag as TRUE and Continue Next Task
T0020	Actions Complete		Go to Step S0020
S0020	Acquire Shared Device, Enable HM and SM	A010	Reset Hold Request and Enable Monitors
		A020	Acquired Shared Devices
		A030	FT1535 Selection
		A040	Change Mode Devices to AUTO/CAS
T0021	BW Pump Fail		Go to Step S0021
T0030	Acquire Failed or Timed Out		Go to Step S0030
T0040	Acquire Complete		Go to Step S0040
T0050A	Task Pointer = 3		Go to Step S0050
S0021	Service Fail	A010	All BW Failed
FAIL	Command to Start Standby Pump Failed		TRUE
S0030	Acquire Failed	A010	Operator Prompt: Acquisition failed for “Device Names”
T0010B	Operator Responded		Go to Step S0010

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0040	Start Flushing 01	A010	Set Task Pointer as 2
		A020	Open Flushing Valves
		A021	FT1535 Selection
		A030	Start UF Feed Pump
		A031	Start Flushing 01 Timer TMR1
		A040	Open UF Product Valves
		A050	Close Flushing Valve DEV5
		EXIT	Continue Next Task
		B010	Display Flushing 01 Timer TMR1
T0050	Actions Complete		Go to Step S0050
S0050	Start Service	A010	Set Task Pointer as 3 and Open UF Service Valve
		A020	FT1535 Selection
		A030	Start UF Feed Pump
		A040	Start Service Timer TMR2
		EXIT	Continue Next. Set Task Pointer as 99
		B010	Display Service Timer TMR2
T9900	Actions Complete		Go to Step S9900
S9900	Complete Service	A009	Raising Backwash Flag
		A010	Stop ALL Pump
		A020	Close ALL Valves
		A030	Disable all HM & SM
		A040	Release Shared Device
		EXIT	Completed
END	Complete Command		TRUE

Table 15: Flushing 01 & Service Sequence

2.4.3 Command 02 – Backwash & Flushing 02

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0000	Start	A010	Enable/Disable Report Parameters and Operational Parameters
T0010	First Pass Check		First Pass = False then S0010
T0011	Backwash & Rinsing can only run by ONE Train		Go to Step S0011
T0020A	First Pass and Task Pointer Check		First Pass = True and Task Pointer != 99 and then S0020
T9900A	First Pass and Task Pointer Check		First Pass = True and Task Pointer = 99 then S9900
S0010	Initialize	A010	Initialize EM
		A020	Initialize Timers
		A030	Set First Pass Flag as TRUE and Set Task Pointer
T0020	Actions Complete		Go to Step S0020
S0011	Backwash Fail	A010	Backwash Failed
FAIL	Command to Start Backwash Failed		TRUE
S0020	Acquire Shared Device, Enable HM and SM	A010	Reset Hold Request and Enable Monitors
		A020	Acquired Shared Devices
		A030	Change Mode Devices to AUTO/CAS
T0030	Acquire Failed or Timed Out		Go to Step S0030
T0040	Acquire Complete		Go to Step S0040
T0050A	Task Pointer = 3		Go to Step S0050
S0030	Acquire Failed	A010	Operator Prompt: Aquisition failed for “Device Names”
T0010B	Operator Responded		Go to Step S0010

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0040	Start Backwash	A010	Set Task Pointer as 2 and Open Backwash Valve DEV5
		A011	Open Air Sourcing Valve DEV7
		A012	Open Backwash Valve DEV6
		A013	Open Backwash Valve DEV4
		A020	Start UF Backwash Pump DEV10 or Standby Pump DEV11
		A021	Start Backwash Timer TMR3
		A030	Stop UF Backwash Pump
		A040	Close Backwash valves & Air Sourcing
		B010	Display Flushing 01 Remaining Timer TMR3
T0050	Actions Complete		Go to Step S0050
S0050	Start Flushing 02	A010	Set Task Pointer as 3 and Open Flushing Valves
		A020	Start UF Feed Pump DEV8 and Flushing 02 Timer TMR4
		A030	Flushing 02 Timer TMR4 Complete
		EXIT	Continue Next. Set Task Pointer as 99
		B010	Display Flushing 02 Timer TMR4
T9900	Actions Complete		Go to Step S9900
S9900	Complete Flushing 02	A010	Stop all Pump
		A020	Close all Valves
		A030	Disable all HM & SM
		A040	Release Shared Device
		EXIT	Completed
END	Complete Command		TRUE

Table 16: Backwash & Flushing 02 Sequence

2.4.4 Command 03 – CEB Backwash

Step / Transition No.	Description	Step / Transition Name	Functional Description
S0000	Start	A010	Enable/Disable Report Parameters and Operational Parameters
T0010	First Pass Check		First Pass = False then S0010
T0011	Backwash & Rinsing can only run by ONE Train		Go to Step S0011
T0011A	Sequence have not run for more than 5 times		Go to Step S0011
T0020A	First Pass and Task Pointer Check		First Pass = True and Task Pointer != 99 and then S0020
T9900A	First Pass and Task Pointer Check		First Pass = True and Task Pointer = 99 then S9900
S0010	Initialize	A010	Initialize EM
		A020	Initialize Timers
		A030	Set First Pass Flag as TRUE and Continue Next Task
T0020	Actions Complete		Go to Step S0020
S0011	Backwash Fail	A010	Backwash Failed
FAIL	Command to Start Backwash Failed		TRUE
S0020		A010	Reset Hold Request and Enable Monitors

Step / Transition No.	Description	Step / Transition Name	Functional Description
	Acquire Device	A020	Acquired Shared Devices
		A030	Continue Next
T0030	Acquire Failed or Timed Out		Go to Step S0030
T0040	Acquire Complete		Go to Step S0040
T0050A	Task Pointer = 3		Go to Step S0050
T0060A	Task Pointer = 4		Go to Step S0060
S0030	Acquire Failed	A010	Operator Prompt: Acquisition failed for “Device Names”
T0010B	Operator Responded		Go to Step S0010
S0040	Start CEB Backwash	A010	Open Backwash valves
		A020	Start UF Backwash Pump
		A021	Start Citric Acid Dosing Pump and NaOCl Dosing Pump
		A022	Start CEB Backwash Timer TMR9
		A030	Stop UF Backwash Pump
		A040	Close Backwash valves
		EXIT	Continue Next
		B010	Display CEB Backwash Timer TMR9
T0050	Actions Complete		Go to Step S0050
S0050	Start CEB Soaking	A010	Start Soaking Timer TMR10
		EXIT	Continue Next
		B010	Display Soaking Timer TMR10
T0060	Actions & TMR10 Complete		Go to Step S0060
S0060	Start CEB Flushing	A010	Open Flushing Valves
		A020	Start UF Feed Pump
		A030	Start CEB Flushing Timer TMR11

Step / Transition No.	Description	Step / Transition Name	Functional Description
		EXIT	Continue Next. Set Task Pointer as 99
		B010	Display CEB Flushing Remaining Timer TMR11
T9900	Actions Complete		Go to Step S9900
S9900	CEB Complete	A010	Stop all Pump
		A020	Close all Valves
		A030	Disable all HM & SM
		A040	Release Shared Device
		A050	Reset CEB Timers
		EXIT	Completed
END	Complete Command		TRUE

Table 17: CEB Backwash Sequence

2.5 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Reporting Parameters				
RP001	Step Counter	Boolean	NA	False
RP002	Backwash Queue	Boolean	NA	False
RP003	PU1535B	Boolean	NA	False
RP004	PU1536A	Boolean	NA	False
RP005	PU1536B	Boolean	NA	False
RP006	Backwash Counter	Boolean	NA	False
RP007	Last Sequence	8-bit Unsigned Integer	NA	1
Operating Parameters				
OP001	Start/Stop/Pause	8-bit Unsigned Integer	NA	0
OP002	BW 0-Stop 1-Pause	8-bit Unsigned Integer	NA	2
OP003	Dosing Pump	8-bit Unsigned Integer	NA	2
Tuning Parameters				
N/A	N/A	N/A	N/A	N/A
Timers				
TMR1	Flushing 01 Timer	Time-out	Second	30
TMR2	Service Timer	Continuous	Minute	45
TMR3	Backwash Timer	Time-out	Second	60
TMR4	Flushing 02 Timer	Time-out	Second	60
TMR5	Air Scouring delay	Time-out	Second	10
TMR6	CEB Backwash	Time-out	Second	30
TMR7	CEB Soaking	Time-out	Minute	15
TMR8	CEB Flushing	Time-out	Minute	15

Table 18: Timers, Reporting, Operating and Tuning Parameters

2.6 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value / Reference
MY_TAG	String	EM_WTP_UF_A
MY_TAG	String	EM_WTP_UF_B
OPR_CMD_MASK	32-bit unsigned integer	2047

Table 19: Configurable Parameters

2.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_WTP_UF	Water Treatment Plant Area: Start up EM

Table 20: Named Set Details

Details of states for NS_WTP_UF named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Service
02	Backwash
03	CEB Backwash

Table 21: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – _EM_WTP_RO_SYS

3.1 Functional Description

EM_WTP_RO_SYS is a class-based equipment module that drives three other sequences to complete reverse osmosis for different products as per the selected jobs. EM_WTP_RO_SYS is derived from the Command Driven equipment module class _EM_CMD16_130 which is defined in a separate document. All devices used in EM shall be acquired in Service Command and It shall be released in Flushing Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as buttons on graphics wherever possible.

3.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions.

Fail Monitor Name	Condition
HM01	Module Ownership Lost
HM02	LT1534 LL
HM03	LT1545 HL
HM04	PSL1541A/B
HM05	PSH1542A/B
HM06	ORP Reading Tripped
HM07	FSL1502 TRUE
HM08	Low Level Chemical Tank
HM09	PT1541A/B HH AND LL
HM10	Conductivity Trip
HM11	Differential Pressure
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.
SM11	Device 11 Failed.

Fail Monitor Name	Condition
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Device 15 Failed.
SM16	Device 16 Failed.
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.

Table 22: Hold Monitors

3.3 Instances

The following Table contains the module instances of the equipment module class.

Instances for Train A

Device Name	Module Class	Instance Name
EM	_EM_WTP_RO_SYS	EM_WTP_RO_A
DEV1	_V12_IP_130_SE	WP1_V1548A
DEV2	_V12_IP_130_SE	WP1_V1549A
DEV3	_V12_IP_130_SE	WP1_V1540A
DEV4	_MVS11_IL_130_SO	WP1_PU1540A
DEV5	_MVS11_IL_130_SO	WP1_PU1540B
DEV6	_MVS11_IL_130_SO	WP1_PU1541A
DEV7	_MRSS11_130_SE	WP1_PU1545A
DEV8	_MRSS11_130_SE	WP1_PU1544A
DEV9	_MRSS11_130_SE	WP1_PU1543A
DEV10	_MRSS11_130_SE	WP1_PU1542A
DEV11	_AI_130_SE	WP1_COT1542A
DEV12	_DI_130_SE	WP1_PSL1541A
DEV13	_DI_130_SE	WP1_PSH1542A
DEV14	_AI_130_SE	WP1_QT1541A
DEV15	_AI_130_SE	WP1_PH1541A
DEV16	_DI_130_SE	WP1_LSLL1546
DEV17	_DI_130_SE	WP1_LSLL1545
DEV18	_DI_130_SE	WP1_LSLL1544
DEV19	_DI_130_SE	WP1_LSLL1543

Table 23: Instances for Train A

<u>Instances for Train B</u>

Device Name	Module Class	Instance Name
EM	_EM_WTP_RO_SYS	EM_WTP_RO_B
DEV1	_V12_IP_130_SE	WP1_V1548B
DEV2	_V12_IP_130_SE	WP1_V1549B
DEV3	_V12_IP_130_SE	WP1_V1540B
DEV4	_MVS11_IL_130_SO	WP1_PU1540C
DEV5	_MVS11_IL_130_SO	WP1_PU1540B
DEV6	_MVS11_IL_130_SO	WP1_PU1541B
DEV7	_MRSS11_130_SE	WP1_PU1545B
DEV8	_MRSS11_130_SE	WP1_PU1544B
DEV9	_MRSS11_130_SE	WP1_PU1543B
DEV10	_MRSS11_130_SE	WP1_PU1542B
DEV11	_AI_130_SE	WP1_COT1542B
DEV12	_DI_130_SE	WP1_PSL1541B
DEV13	_DI_130_SE	WP1_PSH1542B
DEV14	_AI_130_SE	WP1_QT1541B
DEV15	_AI_130_SE	WP1_PH1541B
DEV16	_DI_130_SE	WP1_LSLL1540
DEV17	_DI_130_SE	WP1_LSLL1549
DEV18	_DI_130_SE	WP1_LSLL1548
DEV19	_DI_130_SE	WP1_LSLL1547

Table 24: Instances for Train B

3.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.4.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ
		EXIT	Common Hold Actions
T_H1000	Actions Complete		Go to step H1000
H1000	Hold	A010	Disable Tuning Parameters

Step	Description	Action	Action Description
		A020	Closing All Valves
		EXIT	Enable Valid Command
T_H9900 A	Actions Complete		Go to step H9900
H9900	Common Hold End	A010	Release Shared Devices
		EXIT	Hold Logic Complete Actions
HOLD_E ND	Actions Complete		Hold Complete

Table 25: Hold Sequence

3.4.2 Command 01 – Flushing

Step	Description	Action	Action Description
S0000	Flushing Start	A010	Enable/Disable Tuning Parameters
T0010	Actions Complete		First Pass = False then S0010
T0020A	Actions Complete		First Pass = True and Task Pointer != 99 and then S0020
S0010	Initialize EM	A010	Reset Prompt
		EXIT	Continue Next Task
T0020	Actions Complete		Go to step S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030A	Actions Failed or Time Out		Go to step S0030
T0040	Actions Complete		Go to step S0040
S0030	Operator Prompt	EXIT	Operator Prompt: "Acquisition device failed. Acquire devices again?"
T0020B	Actions Complete		Go to step S0020
S0040	Array 1 Flush	A010	Open array 1 flush valve (DEV1 and DEV2)
		A020	Open RO Feed Pump (DEV4)
		A030	Start Chemical Pumps (DEV7, 8, 9 and 10)
		A040	Open RO High Feed Pressure
		A050	Start Flush Timer (TMR1) and 48 hours Biocide Timer (TMR5)

Step	Description	Action	Action Description
		EXIT	Open DEV3
		B010	Display flush 1 timer
T0030B	Actions Failed or Time Out		Go to step S0030
T0050	Actions Complete		Go to step S0050
S0050	Array 2 Flush	A010	Open array 2 Flush Valve (DEV1)
		A020	Open RO Feed Pump (DEV4) and Close DEV2
		A030	Start Chemical Pumps (DEV7, 8, 9 and 10)
		A040	Start Flush Timer (TMR2)
		EXIT	Close DEV3
		B010	Display flush 2 timer
T0030C	Actions Failed or Time Out		Go to step S0030
T0060	Actions Complete		Go to step S0060
S0060	RO System A	A010	Reset Timer (TMR3) and open DEV1
		A020	Start Chemical Pumps (DEV7, 8, 9 and 10) and RO High Pressure Pump (DEV6)
		EXIT	Service Timer Complete (TMR4)
		B010	Display Service timer
T0030D	Actions Failed or Time Out		TMR4 complete and DEV11 PV value less than set point (TP002) then go to step S0030
T0050B	Actions Complete		TMR4 complete and DEV11 PV value less than set point (TP002) then go to step S0050

Table 26: Flushing Sequence

3.5 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Reporting Parameters				
RP001	PU1540B	Boolean	NA	False
RP002	PU1541A/B	Boolean	NA	False
RP003	PU1545A/B	Boolean	NA	False
RP004	PU1544A/B	Boolean	NA	False
RP005	PU1543A/B	Boolean	NA	False
RP006	PU1542A/B	Boolean	NA	False
RP007	PU1540A/C	Boolean	NA	False
RP008	PU1547A/PU1557B	Boolean	NA	False
Tuning Parameters				
TP002	RO Permeate Conductivity set point	External Reference	NA	WP1_COT1542A/AI1/HI_HI_LIM
TP006	ORP Reading	Floating Point	NA	10
TP007	pH Monitor	Floating Point	NA	10
TP008	Differential Pressure	Floating Point	NA	10
Timers				
TMR1	Flushing 1 Timer	Time_Out	Sec	0
TMR2	Flushing 2 Timer	Time_Out	Sec	0
TMR3	RO Permeate Conductivity timer	Time_Out	Sec	0
TMR4	Service Timer	Time_Out	Sec	0
TMR5	48 hours Biocide	Time_Out	Hour	0
TMR6	1-hour Biocide	Time_Out	Hour	0

Table 27: Reporting, Tuning Parameters and Timers

3.6 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value / Reference
MY_TAG	String	EM_WTP_RO_A
MY_TAG	String	EM_WTP_RO_B
OPR_CMD_MASK	32 bit unsigned integer	3

Table 28: Configurable Parameters

3.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_WTP_RO	RO EM

Table 29: Named Set Details

Details of states for NS_WTP_RO named set are as follows,

Named Set Value	Named Set Name
00	HOLD
01	Flushing

Table 30: Name Set State Details

4 EQUIPMENT MODULE CLASS DESIGN – _EM_WTP_MIXBED

4.1 Functional Description

EM_WTP_MIXBED_SYS is a class-based equipment module that drives three other sequences to complete mixbed system for different products as per the selected jobs. EM_WTP_MIXBED_SYS is derived from the Command Driven equipment module class _EM_CMD16_130 which is defined in a separate document. All devices used in EM shall be acquired in Purging Command and It shall be released in Final Rinse Command. Hold Request shall be enabled for all acquired devices. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as buttons on graphics wherever possible.

4.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions.

Fail Monitor Name	Condition
HM01	Module Ownership Lost
HM02	RO Retention Tank (LT1545) is in low level
HM03	RO Storage Tank Level (LT1546) is in high level
HM04	Demin Storage Tank Level (LT1557) is in high level
HM05	Flow Switch (FSL1502) detect no flow
HM06	Flow Switch (FSL1503) detect no flow
HM07	Acid Measuring Tank (LT1554) is in low level
HM08	Caustic Measuring Tank (LT1553) is in low level
HM09	Acid Bulk Tank (LT1551) is in low level
HM10	Caustic Bulk Tank (LT1552) is in low level
HM11	FT1559 LL
HM12	Demin Conductivity Trip
HM13	FT1559 HH
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.

Fail Monitor Name	Condition
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Device 15 Failed.
SM16	Device 16 Failed.
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.
SM20	Device 20 Failed.
SM21	Device 21 Failed.
SM22	Device 22 Failed.
SM23	Device 23 Failed.
SM24	Device 24 Failed.

Table 31: Hold Monitors

4.3 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	_EM_WTP_MIXBED	EM_WTP_MIXBED
DEV1	_V12_IP_130_SE	WP1_V1552A
DEV2	_V12_IP_130_SE	WP1_V1553A
DEV3	_V12_IP_130_SE	WP1_V1554A
DEV4	_V12_IP_130_SE	WP1_V1555A
DEV5	_V12_IP_130_SE	WP1_V1556
DEV6	_V12_IP_130_SE	WP1_V1557
DEV7	_V12_IP_130_SE	WP1_V1558
DEV8	_V12_IP_130_SE	WP1_V1559
DEV9	_V12_IP_130_SE	WP1_V1550
DEV10	_V12_IP_130_SE	WP1_V1551
DEV11	_V12_IP_130_SE	WP1_V1552B
DEV12	_V12_IP_130_SE	WP1_V1553B
DEV13	_V12_IP_130_SE	WP1_V1554B
DEV14	_V12_IP_130_SE	WP1_V1555B
DEV15	_MVS11_IL_130_SO	WP1_PU1547A

Device Name	Module Class	Instance Name
DEV16	_MVS11_IL_130_SO	WP1_PU1557B
DEV17	_MVS11_IL_130_SO	WP1_PU1557C
DEV18	_MRSS11_130_SE	WP1_PU1558A
DEV19	_MRSS11_130_SE	WP1_PU1558B
DEV20	_MRSS11_130_SE	WP1_PU1559
DEV21	_MRSS11_130_SE	WP1_PU1550
DEV22	_AI_130_SE	WP1_COT1553
DEV23	_DO_130_SE	WP1_SV1552
DEV24	_DO_130_SE	WP1_SV1551

Table 32: Instances

4.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

4.4.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T_H1000	Actions Complete		Go to step H1000
H1000	Hold Service	A010	Stop Pump Actions for Command 1: Service
		A020	Closing All Valves
		EXIT	Enable Valid Command
T_H9900	Actions Complete		Go to H9900
H9900	Common Hold End	A010	Reset Hold Monitor
		A020	Release Shared Devices
		EXIT	Hold Logic Complete Actions
HOLD_END	Actions Complete		Hold Complete

Table 33: Hold Sequence

4.4.2 Command 01 – Purging

Step	Description	Action	Action Description
S0000	Purging Start	A010	Enable Tuning Parameters
T0010	Actions Complete		First Pass = False then S0010
T0020A	Actions Complete		First Pass = True and Task Pointer != 99 and then S0020
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020	Actions Complete		Go to S0020
S0020	Acquire Devices	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030	Actions Failed or Time Out		Acquire Failed then go to S0030
T0040	Actions Complete		Go to S0020
T0050A	Task Pointer =2		Go to S0030
S0030	Operator Prompt	A010	Acquire Failed Prompt (OK)
T0020B	Actions Complete		Go to step S0020
S0040	Check Conductivity	EXIT	Continue Next Task
T0100A	Actions Complete		DEV22 < TP001 then go to S0100
T0200	Actions Complete		DEV22>= TP001 and TASK_PTR.CV' = 1 then go to S0200
T0300A	Actions Complete		DEV22>= TP001 and TASK_PTR.CV' != 1 then go to S0300
S0100	Open Mixbed Valve Service	A010	Open Mixbed Valve Purging (DEV1 and DEV12)
		A020	Starting Mixed Bed Feed Pump (DEV17)
		EXIT	Continue Next Task
T0020C	Actions Complete		DEV22 >= TP001 then go to S0040
T9900A	Actions Complete		RP020 = 2 then END

Step	Description	Action	Action Description
S0200	Timer Conductivity	A010	Reset timer (TMR13)
		A020	Run Timer for Conductivity (TMR13)
T0100C	Actions Complete		DEV22< TP001 then go to S0100
T0300	Actions Complete		DEV22>= TP001 then go to S0300
S0300	Open Mixbed Valve Purging	A010	Open Mixbed Valve Purging (DEV1 and DEV8)
		A020	Start Demin Mixed Bed Feed Pump (DEV17)
		A030	Start Demin Conductivity Timer (TMR1)
		EXIT	Continue Next Task
T0100B	Actions Complete		DEV22 < TP001 then go to S0100
T9900B	Actions Complete		RP020 = 2 then END

Table 34: Purging Sequence

4.4.3 Command 02 – Regeneration

Step	Description	Action	Action Description
S0000	Regeneration Start	A010	Enable Tuning Parameters
T0010	Actions Complete		First Pass = False or then S0010
T0020A	Actions Complete		First Pass = True and Task Pointer != 99 and then S0020
S0010	Initialize EM	A010	First Pass Logic
		A020	Resetting all timers
		EXIT	Continue Next Task
T0020	Actions Complete		Go to S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030	Actions Failed or Time Out		Acquire Failed
T0040	Actions Complete		Task Pointer = 1 then S0040

Step	Description	Action	Action Description
T0050A	Actions Complete		Task Pointer = 2 then S0050
T0060A	Actions Complete		Task Pointer = 3 then S0060
T0070A	Actions Complete		Task Pointer = 4 then S0070
T0080A	Actions Complete		Task Pointer = 5 then S0080
T0090A	Actions Complete		Task Pointer = 6 then S0090
T0100A	Actions Complete		Task Pointer = 7 then S0100
S0030	OAR Acquire Fail	A010	Acquire Failed Prompt (OK)
T0020B	Actions Complete		OAR Responded
S0040	Regeneration	A010	Open Mixbed Valve Backwash (DEV2 and DEV7)
		A020	Starting Mixbed Regen Pump (DEV18 or DEV19)
		A030	Start Backwash Timer (TMR2)
		A040	Stop Mixbed Regen Pump (DEV18 and DEV19)
		A050	Close Mixbed Backwash Valve (DEV2 and DEV7)
		EXIT	Continue Next Task
		B010	Display Backwash timer
T0050	Actions Complete		Go to S0050
S0050	Settle	A010	Close All Valves
		A020	Start Settle Timer (TMR3)
		EXIT	Continue Next Task
		B010	Display Settle timer
T0060	Actions Complete		Go to S0060
S0060	Acid/Caustic	A010	Open Acid/Caustic Suction Valves
		A020	Start Mixbed Regen Pump (DEV18 or DEV19)
		A030	Start Acid/Caustic Suction Timer (TMR4 and TMR5)
		A040	Start Acid Slowrinse Timer (TMR6)
		A050	Start Caustic Slowrinse Timer (TMR7)
		EXIT	Continue Next Task

Step	Description	Action	Action Description
		B010	Display Acid/Caustic timer
T0070	Actions Complete		Go to S0070
S0070	Rapid Rinse	A010	Running Mixbed Rapid Rinse Sequence
		A020	Opening Rapid Rinse Valves (DEV1, 2 and 9)
		A030	Starting Demin Mixed Bed Feed Pump (DEV16 or DEV17)
		A040	Start Rapid Rinse Timer (TMR8)
		A050	Stopping Demin Mixed Bed Pump (DEV16 or DEV17)
		A060	Closing Rapid Rinse Valves (DEV1, 2, 7 and 9)
		EXIT	Continue Next Task
		B010	Display Rapid Rinse timer
T0080	Actions Complete		Go to S0080
S0080	Draindown	A010	Opening Draindown Valves (DEV8 and DEV10)
		A020	Start Draindown Timer (TMR9)
		A030	Closing Mixbed Draindown Valve (DEV8)
		EXIT	Continue Next Task
		B010	Display Draindown timer
T0090	Actions Complete		Go to S0090
S0090	Airmix	A010	Opening Airmix Valves (DEV10 and DEV11)
		A020	Start Airmix Timer (TMR10)
		A030	Closing Mixbed Airmix Valve (DEV10 and DEV11)
		EXIT	Continue Next Task
		B010	Display Airmix timer
T0100	Actions Complete		Go to T0100
S0100	Final Rinse	A010	Opening Final Rinse Valves (DEV1, DEV8 and DEV10)
		A020	Starting Mixed Regen Pump (DEV16 or DEV17)
		A030	Start Final Rinse Timer (TMR12)
		A040	Closing Valve (V1551)
		A050	Stop Demin Mixed Bed Feed Pump (DEV16 or DEV17)
		A060	Close Mixbed Final Rinse Valve (DEV1 and DEV8)

Step	Description	Action	Action Description
		EXIT	Continue Next Task
		B010	Display Final Rinse Timer
T9900	Actions Complete		Go to S9900
S9900	Regeneration Complete	A010	Release Devices
		A020	Set Hold Request
		EXIT	Reset timers and Common Hold Actions
END	Actions Complete		True

Table 35: Regeneration Sequence

4.5 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Operating Parameters				
OP001	Acid Transfer Pump	Boolean	NA	False
OP002	Caustic Transfer Pump	Boolean	NA	False
Tuning Parameters				
TP001	Demin Conductivity Monitor	Floating Point	NA	0
Reporting Parameters				
RP020	Last Command	Floating Point	NA	1
Timers				
TMR1	Purging Timer	Time-out	Mins	0
TMR2	Backwash Timer	Time-out	Mins	0
TMR3	Settle Timer	Time-out	Mins	0
TMR4	Acid Suction Timer	Time-out	Mins	0
TMR5	Caustic Suction Timer	Time-out	Mins	0
TMR6	Acid Slowrinse Timer	Time-out	Mins	0
TMR7	Caustic Slowrinse Timer	Time-out	Mins	0
TMR8	Rapid Rinse Timer	Time-out	Mins	0
TMR9	Draindown Timer	Time-out	Mins	0
TMR10	Airmix Timer	Time-out	Mins	0
TMR11	Mixbed Valve Final Rinse Timer	Time-out	Secs	0
TMR12	Final Rinse Timer	Time-out	Mins	0
TMR13	Denim Conductivity Timer	Time-out	Mins	0

Table 36: Operating, Tuning, Reporting Parameters and Timers

4.6 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value / Reference
MY_TAG	String	EM_WTP_MIXBED
OPR_CMD_MASK	32 bit unsigned integer	7

Table 37: Configurable Parameters

4.7 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_WTP_MIXBED	Mixbed EM

Table 38: Named Set Details

Details of states for NS_WTP_MIXBED named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Purging
02	Regeneration
03	Stop

Table 39: Name Set State Details